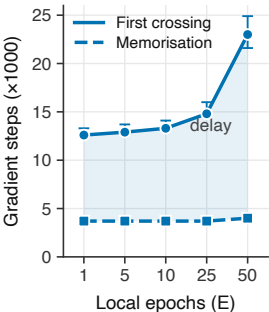
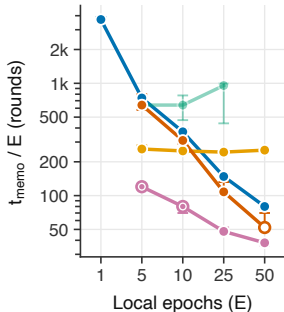


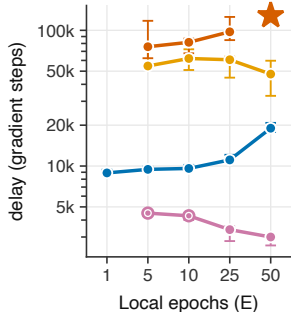
(a) Grokking time vs E (setup A)



(b) Memorisation rounds vs E



(c) Delay vs E



● A: quad-MLP, mod 97, GD

● B: transformer, mod 113

● C: transformer, S_5

● D: quad-MLP, S_5

● E: MLP, MNIST

★ Fixed point (Fig. 6)

✗ Failed to grok